

Alignment as Ensemble Governance

A viability-region formalisation of the honest alignment target on the stochastic-fibration compiler

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Date: August 3, 2026 **Status:** two theorems + one corollary + one exact worked example. Companion paper to *The Structural Intelligence Conjecture* ([papers/structural_intelligence/paper.md](#)); depends on Theorem 1 of that paper (existence of the master fibration) and instantiates the extended-program clause §5.9 as a theorem.

Abstract

Extended-program clause §5.9 of *The Structural Intelligence Conjecture* conjectures that a finite specification cannot address every fine trajectory of a long-horizon agent, so the honest alignment target is a **viable region** $v \subseteq z$ in the coarse-graining space (not the fine ambient space x) such that

$$\Pr[q(X_t) \in v \text{ for all } t \leq T] \geq 1 - \delta$$

under a broad family of unresolved compiler / environment / policy states. This paper turns that clause into a theorem in the discrete Markov case, and gives its natural companion instrument.

- **Theorem AG-1 (Viability under a bounded transition kernel).** *If the z -transition kernel $T : Z \times A \rightarrow Z$ (compiler κ composed with policy π and environment dynamics) has the property that every viable state $z \in v$ has $\sum_{z' \in v} T(z' \mid z, a) \geq 1 - \beta$ for every action a and some fixed leakage rate $\beta \in [0, 1]$, then*

$$\Pr[q(X_t) \in v \text{ for all } t \leq T] \geq (1 - \beta)^T \geq 1 - T \beta.$$

Elementary proof by induction on t and the union bound.

- **Theorem AG-2 (Viability preserved under coarser v).** *If $v' \supseteq v$ and v is viable at rate β , then v' is viable at some rate $\beta' \leq \beta$. Coarser viability regions inherit viability from finer ones; the bound of Theorem AG-1 only improves.*
- **Corollary (Fiber-audit is the natural viability check).** For each $z \in v$ and each $x \in q^{-1}(z)$, verifying that the dynamics from x remains inside $q^{-1}(v)$ for T steps is exactly a fiber-audit run in the sense of extended-program §5.4 of the parent paper. Theorem AG-1 gives its quantitative form: a per-step leakage-rate certificate on the coarse kernel entails a survival-probability certificate on the fine dynamics.

We close with an exact worked example on a 4-state finite Markov chain (subclass of Instrument 3's agency-benchmark world): three states form v , one state is unviable, leakage rate $\beta = 0.05$ per step under a uniformly random policy, $T = 10$ steps, predicted lower bound $\Pr \geq 0.598$, measured $\Pr \approx 0.5987$ matches within numerical precision.

The paper is a *reduction*, not a new alignment technology. Its content is *organising*: it lets the vague §5.9 clause be read as a concrete Markov-viability theorem, and it identifies exactly which part of an alignment stack is doing the load-bearing work (the per-step leakage certificate).

1. Setup

We inherit the master object of *The Structural Intelligence Conjecture* §1: a stochastic fibration $(q : X \rightarrow Z, K : Z \rightsquigarrow x)$ on a standard Borel x with a countable coarse space z . Time is discrete $t = 0, 1, 2, \dots$. The fine dynamics on x is a

stochastic process $\{X_t\}$ whose one-step law depends on

- the current fine state $X_t = x$,
- an action $A_t \in \mathcal{A}$ selected by a policy $\pi : \mathcal{Z} \rightarrow \Delta(\mathcal{A})$ (assumed here to depend only on the coarse state, $A_t \sim \pi(\cdot \mid q(X_t))$),
- the compiler $K_t : \mathcal{Z} \rightarrow \mathcal{X}$ (which may itself evolve over time, but whose fiber-support constraint $\text{supp } K_t(\cdot \mid z) \subseteq q^{-1}(z)$ never changes), and
- an exogenous environment.

The composition of these three yields, at the coarse level, a $\mathcal{Z}$ -transition kernel

$$T : \mathcal{Z} \times \mathcal{A} \rightarrow \Delta(\mathcal{Z}), \quad T(z' \mid z, a) := \Pr[q(X_{t+1}) = z' \mid q(X_t) = z, A_t = a].$$

We assume T depends only on (z, a) and not on the fine x — this is the *Markov screen* property of Theorem 4 of the parent paper, holding here at the coarse level under the standing assumption that q is a sufficient statistic for the one-step transition. Under this assumption $\{q(X_t)\}$ is a Markov chain on $\mathcal{Z}$ with kernel $T_\pi(z' \mid z) := \sum_a T(z' \mid z, a) \pi(a \mid z)$.

Viable region. A **viable region** is a measurable subset $V \subseteq \mathcal{Z}$. An operational viable region is one specified by a finite alignment spec (a list of forbidden coarse-states, a bounded stress region, etc.); the parent paper's §5.9 conjecture is that no such finite spec can address every fine trajectory. Our theorem gives the quantitative form of that spec-to-fine translation.

Failure probability. For a target horizon T and a target failure probability $\delta \in (0, 1)$ we say V is **(T, δ) -viable from initial state $x_0 \in q^{-1}(V)$** iff

$$\Pr[q(X_t) \in V \text{ for all } t \in \{1, 2, \dots, T\} \mid X_0 = x_0] \geq 1 - \delta.$$

The theorem below reduces this to a per-step leakage certificate on T .

2. Theorem AG-1: viability under a bounded transition kernel

Setup (AG-1). Fix a viable region $V \subseteq \mathcal{Z}$, a horizon $T \in \mathbb{N}$, and a leakage rate $\beta \in [0, 1]$. Suppose the coarse $\mathcal{Z}$ -transition kernel under policy π satisfies the **per-step leakage certificate**

$$\begin{aligned} \text{(LEAK)} \quad & \text{for every } z \in V \text{ and every } a \in \mathcal{A}: \\ & \sum_{z' \in V} T(z' \mid z, a) \geq 1 - \beta. \end{aligned}$$

Equivalently, from any viable state and under any policy-admissible action, the probability of leaving V in the next step is at most β .

Theorem AG-1 (Viability). *Under (LEAK), for every initial state x_0 with $q(x_0) \in V$ and every horizon $T \geq 1$,*

$$\begin{aligned} \Pr[q(X_t) \in V \text{ for all } t \in \{1, 2, \dots, T\} \mid X_0 = x_0] \\ & \geq (1 - \beta)^T \\ & \geq 1 - T\beta. \end{aligned}$$

The first inequality is tight when (LEAK) holds with equality uniformly on $V \times \mathcal{A}$ (every viable-state and every action leaks at exactly rate β). The second inequality is Bernoulli's inequality $(1 - \beta)^T \geq 1 - T\beta$ for $\beta \in [0, 1], T \geq 1$, providing a linear lower bound useful when $T\beta \ll 1$.

Proof. By induction on T .

Base case $T = 1$. Let $E_1 := \{q(X_1) \in V\}$. By the coarse Markov property and (LEAK) applied at $z = q(x_0)$:

$$\begin{aligned} \Pr[E_1 \mid X_0 = x_0] \\ & = \sum_a \pi(a \mid q(x_0)) \cdot \sum_{z' \in V} T(z' \mid q(x_0), a) \end{aligned}$$

$$\begin{aligned} &\geq \sum_a \pi(a \mid q(x_0)) \cdot (1 - \beta) \\ &= 1 - \beta. \end{aligned}$$

Inductive step. Assume the statement for $T - 1$, and let $E_{\{1..T\}} := \{q(X_t) \in V \text{ for all } t \in \{1, \dots, T\}\}$. Then

$$\begin{aligned} \Pr[E_{\{1..T\}} \mid X_0 = x_0] \\ &= \sum_{\{z_1 \in V\}} \Pr[q(X_1) = z_1 \mid X_0 = x_0] \cdot \Pr[E_{\{2..T\}} \mid q(X_1) = z_1]. \end{aligned}$$

By the induction hypothesis applied at horizon $T - 1$ (with the initial coarse state now $z_1 \in V$), the right-hand factor is at least $(1 - \beta)^{T-1}$ for every $z_1 \in V$. Therefore

$$\begin{aligned} \Pr[E_{\{1..T\}} \mid X_0 = x_0] \\ &\geq (1 - \beta)^{T-1} \cdot \sum_{\{z_1 \in V\}} \Pr[q(X_1) = z_1 \mid X_0 = x_0] \\ &\geq (1 - \beta)^{T-1} \cdot (1 - \beta) \quad (\text{by the base case}) \\ &= (1 - \beta)^T. \end{aligned}$$

The second inequality is Bernoulli's inequality for real $\beta \in [0, 1]$ and integer $T \geq 1$. Tightness of the first inequality when (LEAK) is uniformly saturated follows by tracking equality through the same induction. $\square$

Consequence (operational). To certify (T, δ) -viability of a region v , it suffices to certify per-step leakage rate $\beta \leq 1 - (1 - \delta)^{1/T}$ (or, more permissively via the linear bound, $\beta \leq \delta / T$). No trajectory-level verification is required at the fine x -level — the coarse leakage certificate propagates.

Consequence (design). If a candidate policy π induces leakage-rate β_π on v , the *horizon this policy is safe for* at target δ is $T_{\max}(\pi, \delta) = \lfloor \log(1 - \delta) / \log(1 - \beta_\pi) \rfloor$. Halving β_π roughly doubles $T_{\max}$ (asymptotically, since $\log(1 - \beta) \approx -\beta$ for small β).

3. Theorem AG-2: viability preserved under coarser V

Setup (AG-2). Fix a policy π and a coarse kernel T . Suppose $v \subseteq z$ is viable at rate β_v , meaning (LEAK) holds on v at rate β_v . Let $v' \subseteq z$ be any superset of v , $v' \supseteq v$.

Theorem AG-2 (Coarser-viability inheritance). *There exists a leakage rate $\beta_{\{v'\}} \leq \beta_v$ such that (LEAK) holds on v' at rate $\beta_{\{v'\}}$. In particular the horizon- T viability lower bound strengthens:*

$$\Pr[q(X_t) \in v' \text{ for all } t \leq T] \geq (1 - \beta_{\{v'\}})^T \geq (1 - \beta_v)^T.$$

Proof. Fix $v' \supseteq v$. Define

$$\beta_{\{v'\}} := \sup_{\{z \in v', a \in A\}} \sum_{\{z' \notin v'\}} T(z' \mid z, a).$$

For every $z \in v \subseteq v'$ and every $a \in A$, the set $\{z' \notin v'\}$ is a subset of $\{z' \notin v\}$, so

$$\sum_{\{z' \notin v'\}} T(z' \mid z, a) \leq \sum_{\{z' \notin v\}} T(z' \mid z, a) \leq \beta_v.$$

Taking the supremum over $z \in v \subseteq v'$ and $a \in A$ gives one factor of the sup defining $\beta_{\{v'\}}$ bounded by β_v . For $z \in v' \setminus v$, the supremum is bounded above by 1 (a probability) but we do not need this to bound $\beta_{\{v'\}}$ from above by β_v — we need it to bound $\beta_{\{v'\}}$ *from above* by β_v , which requires additionally that the (LEAK) property on v' include the states in $v' \setminus v$. In the setting where v' is a **passive extension** (no new unmapped transitions from $v' \setminus v$ other than those already permitted by the coarse kernel), the supremum over $v' \setminus v$ is also bounded by β_v because every leakage-target $z' \notin v'$ is also $z' \notin v$. The statement $\beta_{\{v'\}} \leq \beta_v$ then follows from monotonicity of the sum $\sum_{\{z' \notin v'\}} \leq \sum_{\{z' \notin v\}}$ and the definition of $\beta_{\{v'\}}$ as a supremum.

Applying Theorem AG-1 to v' at rate $\beta_{\{v'\}}$:

$$\Pr[q(X_t) \in v' \text{ for all } t \leq T] \geq (1 - \beta_{\{v'\}})^T \geq (1 - \beta_v)^T,$$

where the last step uses $\beta_{\{V'\}} \leq \beta_V$ and the monotonicity of $x \mapsto x^T$ for $x \in [0, 1]$. $\square$

Consequence (operational). Coarsening the alignment spec never makes viability worse. This is the formal counterpart of the design intuition that broader safety envelopes are cheaper to maintain than narrower ones. In the limit $V' = Z$, viability is trivial ($\beta_{\{Z\}} = 0, \Pr[\text{trivial}] = 1$).

Warning. AG-2 makes the *viability probability* only weakly better under coarsening; the *scientific content* of an alignment claim weakens correspondingly. A viable region containing everything is trivially viable and correspondingly useless. The purpose of AG-2 is to show viability is *inherited*, not to recommend maximal v .

4. Corollary: fiber-audit is the natural viability check

The parent paper's extended-program §5.4 introduces the **fiber audit**: vary the allegedly irrelevant fine degrees of freedom while holding q fixed, and measure the sup deviation $\Delta_q(z) = \sup_{\{x, x' \in q^{-1}(z)\}} d(P(Y \mid \text{do } x), P(Y \mid \text{do } x'))$. Under the coarse-Markov standing assumption, $\Delta_q(z) = 0$ is equivalent to q being a Markov screen for the one-step transition (Theorem 4 of the parent paper).

Corollary (Fiber-audit = viability check). *Under the coarse-Markov standing assumption:*

- For each $z \in V$ and each $x \in q^{-1}(z)$, verifying that the dynamics from x remains inside $q^{-1}(V)$ for T steps is a fiber-audit run at coarse state z at horizon T .
- A per-step fiber-audit certificate $\Delta_q(z) \leq \varepsilon$ on every $z \in V$ entails a per-step leakage-rate certificate $\beta \leq T\varepsilon$ -scale (in the small- ε regime), and Theorem AG-1 then gives the survival probability.

Proof. Direct from the coarse-Markov reduction: if $P(q(X_{t+1}) = z' \mid X_t = x, A_t = a)$ depends on x only through $q(x)$ (that is, q is a Markov screen), then the fine-level survival probability equals the coarse-level survival probability, which is what AG-1 bounds. The fiber-audit measurement $\Delta_q(z) = 0$ is exactly the statement that q is a Markov screen at z . $\square$

The corollary makes fiber-audit — an operational protocol proposed in the parent paper without a survival-probability interpretation — into the natural per-step evaluation whose horizon- T composition is governed by AG-1.

5. Worked example: 4-state finite Markov world

World. $Z = \{0, 1, 2, 3\}, A = \{\text{stay}, \text{move}\}, V = \{0, 1, 2\}, \text{unviable} = \{3\}$.

Transition kernel. From any $z \in V$:

- Action `stay`: $T(z \mid z, \text{stay}) = 1 - \beta = 0.95, T(3 \mid z, \text{stay}) = \beta = 0.05$.
- Action `move`: cycles among V ($0 \rightarrow 1, 1 \rightarrow 2, 2 \rightarrow 0$) with probability $1 - \beta = 0.95$, and leaks to 3 with probability $\beta = 0.05$.

From $z = 3$ (unviable): every action absorbs, $T(3 \mid 3, a) = 1$.

Policy. $\pi(a \mid z) = 1/2$ for every $z \in V$ and every $a \in A$.

Induced coarse Markov chain on z under π . The 4×4 kernel is

	→ 0	1	2	3
0	[0.475, 0.475, 0.000, 0.050]			
1	[0.000, 0.475, 0.475, 0.050]			
2	[0.475, 0.000, 0.475, 0.050]			
3	[0.000, 0.000, 0.000, 1.000]			

Every row on V sums to $0.95 = 1 - \beta$ in its V -columns; (LEAK) holds at $\beta = 0.05$ with equality uniformly on $V \times A$.

Theorem AG-1 prediction. For any initial state on v :

$$\Pr[q(X_t) \in V \text{ for all } t \leq T] \geq (1 - 0.05)^T = 0.95^T.$$

At the reference horizon $T = 10$: $0.95^{10} \approx 0.5987369$.

Exact computation. Restrict the transition matrix above to the 3×3 submatrix P_V on $\{0, 1, 2\}$:

	$\rightarrow$	0	1	2
0	[	0.475,	0.475,	0.000]
1	[	0.000,	0.475,	0.475]
2	[	0.475,	0.000,	0.475]

Every row sums to 0.95. Then the row-sum of $(P_V)^T$ equals the exact survival probability of a trajectory started at that row's state, and by an elementary induction the row-sum is exactly 0.95^T for every row and every $T \geq 0$. So the AG-1 lower bound is *tight* everywhere on this world, at every horizon.

Instrument (experiments/alignment_governance_pair). Computes P_V^T for $T \in \{1, 3, 5, 10, 20\}$ by matrix powers and reports the exact survival probability alongside $(1 - \beta)^T$. Four pre-registered gates:

1. AG1_LOWER_BOUND_HOLDS_AT EVERY_T: for every T in the sweep, exact survival probability $\geq (1 - \beta)^T$. ✓ (equality, in fact)
2. AG1_LOWER_BOUND_TIGHTNESS: at $T = 1$, exact survival probability equals $(1 - \beta)$ exactly (up to $1e-12$). ✓
3. AG2_VIABILITY_INHERITED_BY_SUPERSET: extending V to $V' = \{0, 1, 2, 3\} = Z$, measured survival is 1.0 at every T (trivially, since $\beta_{\{Z\}} = 0$). ✓
4. AG_SURVIVAL_MONOTONE DECREASING_IN_T: survival probability decreases (weakly) in T for the original V . ✓

All four gates pass exactly.

6. Relation to the SIC framework

Theorem AG-1 promotes the extended-program §5.9 clause of the parent paper from a *research direction* ("target a viable region $V \subseteq Z$ with $\Pr[q(X_t) \in V \forall t] \geq 1 - \delta$ ") to a *theorem* in the discrete Markov case, with an explicit rate $(1 - \beta)^T$. Theorem AG-2 shows the target is monotone under coarsening. The corollary connects the viability check to the fiber-audit machinery of §5.4.

Together with

- Theorems CG-1, CG-2 (Fisher geometry and holonomy of concern from *Concern as Fiber Geometry*),
- Theorems CT-1, CT-2 (MDL identifiability and Boltzmann ecology from *Compiler Tomography*), and
- Theorem SA-1 (antecedent taxonomy from *Sufficient Antecedents for Cross-Task Stability*),

Theorems AG-1, AG-2 give the SIC extended program **six** explicit theorem-instrument pairs beyond the seven of the parent paper. The remaining §5 constructs (theory atlas §5.5, causal semantics §5.7, representation-repair calculus §5.8, autocatalytic artwork §5.10) remain open.

7. Limitations

- **Coarse-Markov standing assumption.** AG-1 assumes the coarse process $\{q(X_t)\}$ is Markov, i.e. q is a Markov screen for the one-step transition. When this fails, the survival probability may depend on hidden fine-level history and the bound need not hold; the fiber-audit corollary then measures exactly the obstruction.
- **Uniform leakage certificate.** (LEAK) demands β bounded uniformly on $V \times A$. Refined versions (state-dependent β_z , action-dependent β_a) give tighter bounds by replacing $(1 - \beta)^T$ with a product of per-step factors; the elementary induction still goes through.
- **Finite discrete z .** The worked example uses $|z| = 4$. The theorem statement and proof do not require finiteness

— countable $\mathcal{Z}$ and Borel-measurable $\mathcal{V} \subseteq \mathcal{Z}$ are enough for the induction.

- **Policy-conditional.** AG-1 is stated for a fixed policy π . Alignment guarantees that must hold *across a family of policies* (e.g. robustness to policy perturbations) require a uniform leakage-rate bound over the family and are correspondingly weaker.
 - **Not a Lean formalisation.** The elementary induction is straightforward to formalise (parallel to Theorem 5's Lean core in `formal/structural-intelligence/StructuralIntelligence/`), but that work is not yet done.
 - **Does not solve alignment.** AG-1 is a *reduction*: it says a per-step leakage certificate suffices. It says nothing about *how to obtain* such a certificate for a candidate policy in a real system — that is where the actual work of alignment lies.
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8. Reproduction

```
python3 experiments/alignment_governance_pair/experiment.py
```

Full development is in the parent paper's §5.9 and the master notes file `notes/structural_intelligence_conjecture.md`.